

Day 4 — Build a Tithi Calendar

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Each student (or pair) will build a tithi calendar for the current calendar month — labelling each date with its tithi name, pakṣa, and expected Moon shape.

Materials

- One printed page of a current-month *pañcāṅga* per pair (Drik Panchang, Mypanchang, or any standard source). Confirm the source and timezone before printing.
- A blank monthly calendar grid (printed handout — see activity-01)
- Coloured pencils or markers (two colours — one for *śukla*, one for *kṛṣṇa*)
- Notebooks

Vocabulary reused today

- *tithi*, *pakṣa*, *śukla pakṣa*, *kṛṣṇa pakṣa*, *amāvasyā*, *pūrṇimā* (Days 1–3)
- All 15 tithi names (Day 3)

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What tithi was it?*” — 2 students share.
- Hand back yesterday’s formative quizzes briefly (90 sec) — call out the most common miss; correct it once.
- “*Today the puzzle is concrete: what’s the tithi for every date this month?*”

Core (15 min)

1. **Show one sample row.** Project or photocopy a *pañcāṅga* entry for any one date. Walk through:

- Date: e.g. “May 22”
- Pakṣa: e.g. *kṛṣṇa*
- Tithi at sunrise: e.g. *caturthī*
- Tithi end-time: e.g. 14:37 (after which *pañcamī* begins)

“*So if you went out at 8 p.m. on May 22, the tithi would already be pañcamī, not caturthī. The tithi listed at the top of a calendar date is the one current at sunrise.*”

2. **The pattern of a month — anticipated.** Most months have:

- A *śukla pratipadā* (the day after *amāvasyā*)
- A *pūrṇimā* (in the middle)

- A *kṛṣṇa pratipadā* (the day after *pūrṇimā*)
 - A *kṛṣṇa amāvasyā* (close to the end)
 - 28–31 dated tithis (because of *kṣaya* / *vṛddhi* — see Day 3)
3. **Anatomy of a tithi calendar — the activity sheet.** Walk through the blank grid:
- One row per date
 - Columns: Date | Day of week | Pakṣa | Tithi name | Tithi end-time | Expected Moon shape
4. **One festival hook.** Find one festival this month from the *pañcāṅga* and announce it: “*On dvitīyā of śukla pakṣa this month, there’s a regional festival called X — let’s see if we agree with the date.*”

Activity (20 min) — Build the calendar

This **IS** Activity 01 — see `activities/activity-01-build-tithi-calendar.md` for the full set of instructions.

In pairs: 1. Receive one printed *pañcāṅga* page for the current month. 2. Fill in the blank calendar grid for every date of the month. 3. Colour code: *śukla pakṣa* in yellow, *kṛṣṇa pakṣa* in grey/blue. 4. Sketch a small Moon shape for each date (or every 3rd date if running short on time). 5. Circle any *amāvasyā* and *pūrṇimā*.

Walk around. The most common error: students copy the wrong column from the *pañcāṅga*. Have them double-check 2–3 entries with you.

Wrap-up (5 min)

- Pairs hold up their finished calendars briefly. The teacher takes a quick scan.
- **Exit question (1 min, oral):** “Which date this month is *pūrṇimā*?” Class answers in chorus.
- **Preview Day 5:** “Tomorrow we name the 12 lunar months and link them to festivals.”

Student-facing content

Today you built a tithi calendar.

A tithi calendar links each date in a calendar month to: - the *pakṣa* it falls in - the *tithi* current at sunrise - the time the tithi ends - the expected shape of the Moon

Two patterns to notice: - There are usually 1 *amāvasyā* and 1 *pūrṇimā* per calendar month — sometimes 0 or 2 depending on month length. - A few tithis will have unusual entries — a *kṣaya* (skipped) or *vṛddhi* (spanning two days). Note them; they are the system working as designed.

Homework

- Continue the Moon diary. Compare last night’s sketch with the calendar you built today — does the shape match the predicted shape for that tithi?
- For Day 5: be ready to name **one festival in your family** (or community) that you know is tied to a specific tithi.

Differentiation

- **One tier up:** Look at the *pañcāṅga* for the month BEFORE this one. Find a tithi that was *kṣaya* (skipped). Explain in writing what happened on that date — the previous tithi was running at sunrise, then the next tithi started after sunrise and ended before the next sunrise. So the “skipped” tithi never had its own sunrise.
- **One tier down:** Fill in the date column and the tithi-name column only. Skip the end-times and the Moon shapes for this lesson.

Teacher notes

- **Choose ONE *pañcāṅga* source for the entire class.** Different *pañcāṅgas* may use different sunrise locations or computational systems and give slightly different tithi names for borderline dates. Pick one (Drik Panchang is well-known; Mypanchang.com is also reliable) and stick to it for the whole module.
- Some calendars list two tithis on a single date (e.g. “Caturthī upto 14:37, then Pañcamī”). This is normal. Have students enter the tithi at sunrise, then note the transition.
- The colour-coding (yellow for *śukla*, blue/grey for *kṛṣṇa*) makes the calendar legible at a glance. Insist on it.
- Bring a backup: if printing fails, project the *pañcāṅga* for the whole class to read together. Less efficient but workable.

Citation(s) used in this lesson

- The class *pañcāṅga* source (named on the board): use this consistently. The data is computed using the same astronomical principles described in the *Sūrya-siddhānta* — paraphrased only; no verse cited.